Short review on *Deep neural networks for improved, impromptu trajectory tracking of quadrotors*

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1. What is the paper about?

It is difficult for traditional, simple controllers to track the trajectory for quadrotors because of non-linear dynamics, difficulties in qualifying uncertainty in the turn-rate-to-trust map, time delays and shortcomings of classical controllers. This paper proposes a deep neural network (DNN) based control system by posing a DNN before the original feedback control loop so that the system can learn past flight experience. After training offline from the past flight experience, tracking errors can be reduced by around 40~50% with no prior knowledge. Moreover, the system can track the trajectory on the "fly-as-you-draw" APP in the real-time. In conclusion, this paper proposed a pre-posed DNN control methods with current state feedback and future desire state input to improve trajectory tracking.

2. What approach did it take?

To guarantee the stability of the system, DNN is pre-posed outside the feedback control loop and send reference signal to the control loop 10 times slower than the loop itself. Before introducing the technical detail, the notation should be given: c is the current, d is the desired and r is the reference. To train the DNN, approximately 10,000 raw data pairs (*a*, *b*) are collected for training, where $a = \{\mathbf{s}^*_{c,t}, \mathbf{s}^*_{c,t+\Delta_1+1}, ..., \mathbf{s}^*_{c,t+\Delta_L+1}\}, b = \mathbf{s}^*_{d,t}$ and $\mathbf{s} = \{x, y, z, \dot{x}, \dot{y}, \dot{z}, \phi, \theta, \psi, p, q, r, \ddot{z}\}$. The neural network can be written as $f : \{\mathbf{s}_{c,t}, \mathbf{s}_{d,t}, ..., \mathbf{s}_{d,t_L}\} \mapsto \mathbf{s}_{d,t}$. Three configuration are investigated:

1. DNN with future desired states and the current state feedback: $L + 1$ sets of $\{\dot{x}, \dot{y}, \dot{z}, \phi, \theta, \psi, p, q, r, \ddot{z}\ \}$ and $\{\mathbf{p}_{d,t+\Delta_1} - \mathbf{p}_{c,t}, ..., \mathbf{p}_{d,t+\Delta_L} - \mathbf{p}_{c,t}\ \}$, where $\mathbf{p} = (x, y, z)$
2. DNN with future desired states and without the current state feedback: It is similar to Configuration 1 with only $\mathbf{s}_{c,t}$ and $\mathbf{p}_{c,t}$ being replace by $\mathbf{s}_{d,t-1}$ and $\mathbf{p}_{d,t-1}$.
3. DNN without future desired states and with the current state feedback: only $\mathbf{s}_{c,t}$.

After this, it is reasonable why the line $\mathbf{s}_c$ in Fig.3 is dashed. Six DNNs are trained for the difference in getting parameters. More details about DNN training can be found in the paper.

3. How could this work be strengthened/improved?

Dynamic Lab has tried multiple methods in improving the performance of quadrotor trajectory tracking from modification of the controller, add-on from DNN, transfer learning, etc (Zhou et al. 2019; Pereida and Schoellig 2019; Pereida, Helwa, and Schoellig 2018; Zhou et al.; Pereida, Duivenvoorden, and Schoellig 2017; Zhou, Helwa, and Schoellig 2018, 2017; Greeff and Schoellig 2018). As for my advice, I want to propose three points:

1. (Li et al. 2017) used six separate networks. It would break the relationship between the system parameter inside. Inspired by (Li et al. 2019; Sun et al. 2019), one network with six branches can be trained. The branch can share the weight in some way so that the dependency and in-dependency can be retained at the same time.
2. In Fig. 6 in (Li et al. 2017), it can be found that the error increases while the speed is larger. Inspired by (Huang et al. 2017; He et al. 2016), it is different to detect different objects in different sizes. The smaller the object is, the harder to detect it. Thus, the most direct idea is to train three separate networks to deal with different situations. Or we can use a network with multiple outputs. The signal at different speeds can be output in different stages. If the speed is low, the reference signal can be output at the shadow layer and vice versa.
3. The relationship between the number of layers and the error is not given. According to the knowledge in deep learning, the deeper the network, the higher the capacity of the network. I don't know whether it is true in this. But I think if it works, we can make the network deeper.
4. (Li et al. 2017) mentioned limited computational resource. For the first problem, mean teacher (Tarvainen and Valpola 2017) and model distilling (Hinton, Vinyals, and Dean 2015) can decrease the size of the model to decrease the computation cost.
5. Besides the common DNN, control with reinforcement learning can be tried (Xie et al. 2018).

4. State what research problems you are interested in

Control with graph neural network; Robotics with reinforcement learning; Robot perception; Robot Navigation; Embedded deep learning.

5. Briefly state why you want to pursue a graduate degree

After three research experiences, I know it is indeed a tough job doing research. One has to be patient and rigorous. But I think I am a qualified candidate who has possessed the above-mentioned characteristics in addition to my academic background and also full of passion to change the world.

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